

Alexandre Ait Ettajer

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SUMMARY

Software engineer with 3+ years of experience in systems development across frontend and backend. Skilled in designing scalable architectures, leading end-to-end projects, and mentoring junior engineers. Focused on backend systems, distributed computing, and computational infrastructure. Passionate about solving challenging problems and continuously deepening core CS knowledge.

SKILLS

Languages: C++, JavaScript, MATLAB, Rust, Python, TypeScript, Fortran

Frameworks/Libraries: Node.js, LAPACK, PyTorch, NumPy, Matplotlib

Tools/Platforms: Docker, GitHub Actions, Git, Perforce

Collaboration: Mentorship, Cross-functional Communication, Stakeholder Alignment, Technical Presentations

EXPERIENCE

The MathWorks, Inc.

Natick, MA (Hybrid)

Software Engineer

April 2025 - Present

- Drove low-level performance improvements in MATLAB visualizations by redesigning client-server messaging, reducing latency and simplifying architecture
- Replaced legacy C++ layout/rendering with JavaScript canvas, enabling smooth client-side interactions without backend roundtrips
- Designed tabular data support for core graphics (Bar, Area, Histogram), integrating SQL/Excel-style datasets
- Created reusable data abstractions to generalize heterogeneous input handling across graphics objects
- Mentored junior engineers and communicated technical direction to senior managers across cross-functional teams
- Contributed to visual tools used by tens to hundreds of thousands of engineers and students worldwide

Senior Associate Software Engineer

April 2024 - April 2025

- Developed 3D data annotations using MATLAB UI and JavaScript rendering, creating infinite plane annotations
- Designed and implemented MATLAB's first hierarchical data viz (Sunburst Chart); led UX and design reviews
- Diagnosed and resolved backend theming and serialization bugs, ensuring compatibility across releases

Associate Software Engineer

Apr 2023 - Apr 2024

- Prototyped small-scale features, gained deep understanding of C++/XML build systems

EDG Associate

May 2022 - Apr 2023

- Solved customer issues with code generation and completed dev rotations

Sandia National Laboratory Intern

Albuquerque, NM (Remote)

Science Undergraduate Laboratory Intern

May 2021 - May 2022

- Automated visualization of parameter effects on HPC performance for MHD fluid simulations
- Presented findings to senior researchers and contributed to internal proposal pipeline

PROJECTS

Canticle for the Doomed — itch.io

Browser-based rogue-like with procedural generation, built during 7-day game jam

TypeScript, Docker, GitHub Actions

Co-developed gameplay and built hot-reload deployment pipelines

EDUCATION

M.S. in Computer Science

GPA: 3.56

The University of Texas at Austin

Austin, TX (December 2025)

Relevant Coursework: Advanced OS, Distributed Computing, Deep Learning, Quantum Computing, NLP, RL

B.S. in Computer Science and B.S. in Aeronautical Engineering

GPA: 3.93, Summa Cum Laude

Rensselaer Polytechnic Institute

Troy, NY (May 2022)

Capstone Projects: Operating Systems (CS), Numerical Solutions for Wave Equations (AE)